May 2025

B. Tech. (CE & CE (Hindi)) (Fourth Semester)

Design and Analysis of Algorithms

(PCC-CS-404)

Time : 3 Hours] [*Maximum Marks : 75*

Note : It is compulsory to answer all the questions (1.5 marks each) of Part A in short. Answer any *four* questions from Part B in detail. Different sub-parts of a question are to be attempted adjacent to each other.

Part A

1. (a) Arrange the following complexity in increasing order :

– n, $\log n$, n^3, $n^2 \log n$, $1/\log n$, 2^n. **1.5**

(b) Differentiate Divide and Conquer and Dynamic Programming algorithm design technique. Justify with appropriate example. **1.5**

(c) Explain Dominance rule with example. 1

(d) Define principle of optimality. In whic technique is it used ? 1.

(e) Which of the following sorting algorithm ar stable : insertion sort, merge sort or quicl sort ? And how ? 1.5

(f) What is the use of approximation algorithms ? 1.5

(g) What are implicit and explicit constraints in backtracking methods ? 1.5

(h) What is least cost search in branch and bound ? Write the equation for the same. 1.5

(i) Differentiate between P, NP, NP-hard and NP-Complete problems. 1.5

(j) What are deterministic and nondeterministic algorithms ? 1.5

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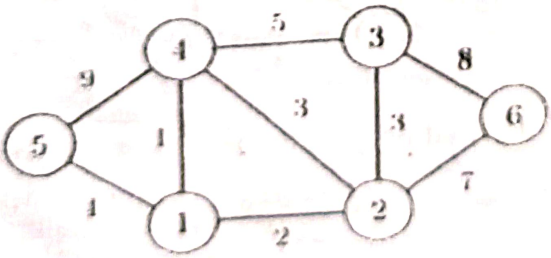
Part B

2. (a) Explain master method for computing complexity of recurrence relations. Solve the following recurrences : 7.5

(i) $T(n) = 3T(n/4) + n \log n$

(ii) $T(n) = T(\sqrt{n}) + n.$

(b) Explain Prim's algorithm. Apply Prim's algorithm to obtain a minimum cost spanning tree for the given graph : 7.5



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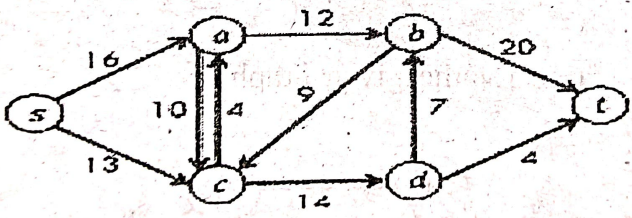
3. (a) Find the product of these two matrices using Strassen's matrix multiplication method. Derive the time complexity of Strassen,s matrix multiplication :

$$A = \begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix}, \quad B = \begin{bmatrix} 8 & 4 \\ 6 & 2 \end{bmatrix}.$$ 7.5

(b) What are residual networks, augmenting paths, and cuts of flow network ? Describe Ford Fulkerson algorithm and solve the following graph for maximizing the network flow : 7.5



4. (a) Find all possible topological sorting orderings of a given Directed Acyclic Graph (DAG) starting from node *a*. 7.5

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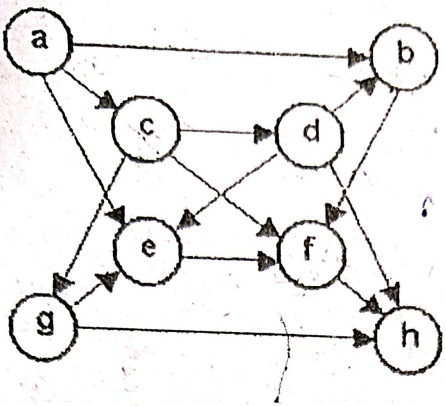

(b) Explain job sequencing with deadlines. Generate a solution using the same for the following jobs : 7.5

Job	A	B	C	D	E
Profit	100	19	27	25	15
Deadline	2	1	2	1	3

5. (a) Explain N-queens problem using backtracking. 7.5

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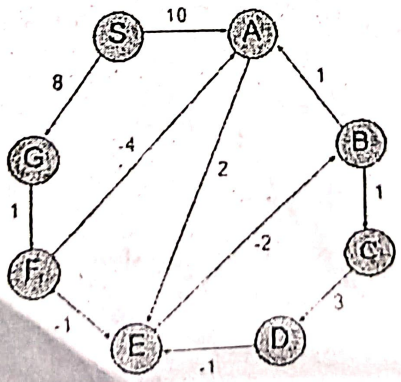
(b) Define branch and bound strategy. Solve the following 0/1 Knaphsack problemusing branch and bound strategy with knapsack weight equals to 15. 7.5

Profit	10	10	12	18
Weight	2	4	6	9

6. (a) Explain Dijkstra algorithm. How is it different from Bellman Ford Algorithm ? Apply the appropriate single shortest path algorithm for the following graph : 7.5



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(b) Write an algorithm for Randomized Quick Sort. Find its worst case complexity. 7.5

7. Write notes on the following :

(a) Las Vegas and Monte Carlo algorithm 5

(b) Hamiltonian Cycle 5

(c) Floyd-Warshall algorithm. 5

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